

- $\frac{I_1}{I_2} = 100^{\frac{1}{3}(m_2 - m_1)} \quad \left| \quad I_i = \frac{L_i}{4\pi D_i^2} \right.$

$$I_2 = \frac{L}{4\pi (10)^2} \text{ (absolute)}$$

$$\frac{L/4\pi D}{L/4\pi (10)^2} = 100^{\frac{1}{3}(M - m)} \quad *$$

$$\left[\begin{array}{l} m - M = 5 \log(D) - 5 \\ \rightarrow \text{add extinction} \end{array} \right.$$

$$m_{\text{obs}} = m_{\text{true}} + A_\lambda$$

$$A_\lambda \sim \frac{1}{\lambda^\alpha}, \quad A_\lambda \sim \frac{1}{\lambda} \quad \text{or} \quad \frac{1}{4} \quad \text{etc.}$$

* Diff. Extinction: $E(B-V) = A_B - A_V = \frac{1}{\lambda_B^\alpha} - \frac{1}{\lambda_V^\alpha}$
A.1

$$R = \frac{A_V}{A_B - A_V} = \frac{A_V}{A_B - A_V}$$

- $A_\lambda \propto 1/\lambda^\alpha$

$$A_\lambda \propto 1/\lambda^\alpha \Rightarrow A_{\lambda_B} = \frac{\alpha_B}{\lambda_B^\alpha} \Rightarrow$$

$$R = \frac{A_V}{A_B - A_V} = \frac{1}{\left(\frac{A_B}{A_V} - 1\right)} = \frac{1}{\left(\frac{\lambda_V^\alpha}{\lambda_B^\alpha} - 1\right)}$$

$A_{B \& V} \quad \alpha_B = \alpha_V \Rightarrow$

$$\lambda_V = 4300 \text{ \AA} \quad \lambda_B = 5500 \text{ \AA}$$

$$R_{ISM} \approx 3$$

$$A_\lambda \sim \frac{1}{\lambda}$$

$$R = 0.596$$

B.1 | 0.5V: $A_V = 0.65 \text{ mag} \quad | \quad A_\lambda \propto \lambda^{-1.2} \quad *$

$$\left\{ \begin{array}{l} \lambda_B = 440 \text{ nm} \\ \lambda_V = 550 \text{ nm} \end{array} \right.$$

$$(M_B)_0 = 5.44$$

$$\bullet A_B - A_V = A_B \left(1 - \frac{A_V}{A_B} \right) = A_B \left(1 - \left(\frac{\lambda_V}{\lambda_B} \right)^{1.2} \right) = A_B \alpha$$

$$A_B - A_V = A_B \alpha$$

$$A_B - A_B \alpha = A_V$$

$$A_B (1 - \alpha) = A_V$$

$$A_B = \frac{A_V}{1 - \alpha} = \frac{A_V}{1 - \left(1 - \left(\frac{\lambda_B}{\lambda_V} \right)^{1.2} \right)} = \left(\frac{\lambda_V}{\lambda_B} \right)^{1.2} A_V$$

$$A_B = \left(\frac{\lambda_V}{\lambda_B} \right)^{1.2} A_V = 0.85 \text{ mag} \sim \frac{1}{\lambda^{1.2}}$$

$$(6) (i) (L_{obs})_B = 1.2 \times 10^4 L_\odot$$

$$\bullet m_{obs} - m_{true} = A_\lambda$$

$$\left(\frac{L_2}{4\pi D_2^2} \right) = 100^{\frac{1}{5}(m_2 - m_1)} \quad \left(\frac{L_2}{4\pi D_2^2} \right) \quad \begin{matrix} \uparrow & \uparrow \\ m_{obs} & m_{true} \end{matrix} \quad \leftarrow$$

$$\frac{L_{true}}{L_{obs}} = 100^{\frac{1}{5}(m_{obs} - m_{true})} = 10^{\frac{2}{5} A_\lambda}$$

$$L_{true} = L_{obs} \cdot 10^{\frac{2}{5} A_\lambda}$$

$$L_{obs}^B = 1.2 \times 10^4 L_\odot$$

$$\Rightarrow L_{true} > L_{obs}$$

$$\bullet m - M = 5 \log(D) - 5$$

$$(iii) \frac{L_2}{4\pi D_2^2} = 100^{\frac{1}{5}(m_2 - m_1)} \quad \left(\frac{L_2}{4\pi D_2^2} \right)$$

absolute Mag: $(M_B)_\odot : D_2 = 10$
 $m_1 \rightarrow (M_B) : D_1 = 10$

$$true \quad (M_B) \quad (M_B)$$

$$\frac{(L_S)_B}{L_{B0}} = 100^{\frac{5}{8}(M_{B0} - M)}$$

$$\log\left(\frac{(L_S)_B^{\text{true}}}{L_0}\right) = \frac{2}{5}(M_{B0} - \hat{M})$$

$$\hat{M} = -\frac{5}{2} \log(L_B^{\text{true}}) + M_{B0}$$

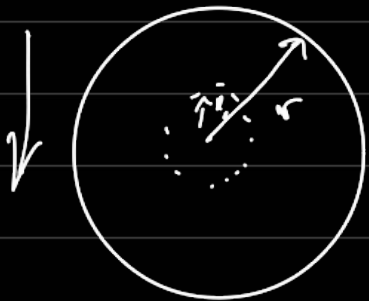
⑥ tang. velocity: (baria)

$$\rightarrow V_t (\text{km s}^{-1}) \approx 4.74 d (\text{kpc}) \mu (\text{mas arcsec yr}^{-1})$$

$$d = \frac{1}{\theta''} \leftarrow$$

$$\textcircled{c} \underline{[S_*]} = \frac{N_*}{\Delta t}, \quad n_H$$

$$\underline{n_H} = n_0 \left(\frac{r}{R_0}\right)^{-2/3} *$$



R_0 : const.
 $n_0 = R_0$

$$n_H(R_0) = n_0$$

R_S : Strömgren radius: $\frac{N}{2} kT \sim \frac{1}{2} m V^2$

Ionization: $N_* = S_* \Delta t$

$$N_* = n_H \text{Vol} : *$$



ΔR

$$\text{Vol} = 4\pi r^2 \Delta r *$$

2D:



$$2\pi a$$

$$\downarrow \Delta n$$

$$\text{Vol} = \Delta a 2\pi a$$

$$V_r = n_H u \pi r^2 \Delta r = S \Delta t$$

Recalls: $R = n_p \alpha_{rec}$ $\left(\frac{1}{\text{time}} : \begin{array}{c} * \\ \text{single} \\ \text{electron} \end{array} \text{ to recombine} \right)$

$$dR_{\text{tot}} = \underbrace{N_e}_{\sum} (R) = n_e (u \pi r^2 dr) n_p \alpha_{rec}$$

$$N_e = n_e (\text{Vol})$$

$$\begin{aligned} \underline{R_{\text{tot}}} &= \int_0^{R_s} \underset{\uparrow}{n_e} \underset{\uparrow}{n_H} u \pi r^2 d_{\text{rec}} dr \\ &= \int_0^{R_s} n_H^2 u \pi r^2 d_{\text{rec}} dr \\ &= \int_0^{R_s} \left(n_0 \left(\frac{r}{R_0} \right)^{-2/3} \right)^2 u \pi r^2 d_{\text{rec}} dr \\ &= \int_0^{R_s} n_0^2 \left(\frac{r}{R_0} \right)^{-4/3} u \pi r^2 d_{\text{rec}} dr \\ &= \int_0^{R_s} n_0 R_0^{4/3} \frac{r^2}{r^{4/3}} u \pi d_{\text{rec}} dr \\ &= u \pi d_{\text{rec}} n_0 R_0^{4/3} \int_0^{R_s} \frac{r^2}{r^{4/3}} dr \\ &= u \pi d_{\text{rec}} n_0 R_0^{4/3} \int_0^{R_s} r^{2/3} dr \\ &= u \pi d_{\text{rec}} n_0 R_0^{4/3} \left(\frac{3}{5} R_s^{5/3} \right) \end{aligned}$$

$$R_{\text{tot}} = \frac{12 \pi}{5} d_{\text{rec}} n_0 R_0^{4/3} R_s^{5/3}$$

* Now eg: was when $R_{\text{tot}} = S$

$$R_s^{5/3} \frac{12 \pi}{5} d_{\text{rec}} n_0 R_0^{4/3} = S$$

Solve for R_s :

B.1 Mock:

(i) They don't give $n_H = n_H(r)$: assume n_H const.

Recomb: $R = \alpha_{rec} n_p$ (single)

$$dR_{tot} = \underbrace{n_e (dVol)}_{N_e} d_{rec} n_p$$



$$\bullet dVol = 4\pi r^2 dr$$

$$dR_{tot} = n_e 4\pi r^2 dr d_{rec} n_p$$

$$\bullet n_e = n_p = n_H:$$

$$\int dR_{tot} = R_{tot} = \int_0^{R_s} n_H^2 4\pi r^2 d_{rec} dr$$

* Assume n_H and d_{rec} are const.

$$\begin{aligned} R_{tot} &= n_H^2 4\pi d_{rec} \int_0^{R_s} r^2 dr \\ &= 4\pi n_H^2 d_{rec} \left(\frac{1}{3} R_s^3 \right) \\ &= \left(\frac{4\pi}{3} R_s^3 \right) n_H^2 d_{rec} \end{aligned}$$

$$\text{* Final: } R_{tot} = S_*$$

solve for R_s :

$$\begin{aligned} R_{tot} &= n_e (Vol) d_{rec} n_p \\ &= n_e \left(\frac{4\pi}{3} R_s^3 \right) d_{rec} n_p \end{aligned}$$

$$\bullet \text{Velocity: } [V] = \frac{D}{t} = \underline{5 \text{ km/s}}$$

$$\underline{D} = VT$$

$\approx$
const.

$$N_\gamma = S_\star \Delta t$$

Const.

B-11 (b) Plug in

① $\lambda = 0.503 \mu\text{m}$

$$L = \frac{\Delta E}{\Delta t}$$

$$(E_1) = \frac{hc}{\lambda} \leftarrow \text{single}$$

$$N_\gamma = S_\star \Delta t$$

$$\Delta t = \frac{N_\gamma}{S_\star}$$

$$(L_1) = \frac{hc S_\star}{\lambda N_\gamma}$$

$$L_{\text{tot}} = N_\gamma L_2 = \frac{hc S_\star}{\lambda} \quad (\text{Plug in})$$

(Last parts) (d-e) (Both)

* Oxygen collisions in ISM:

• within ISM

→ H, e, p ...

→ O, S, N

→ Ionized Oxygen: in H II

High energy photons also ionize O

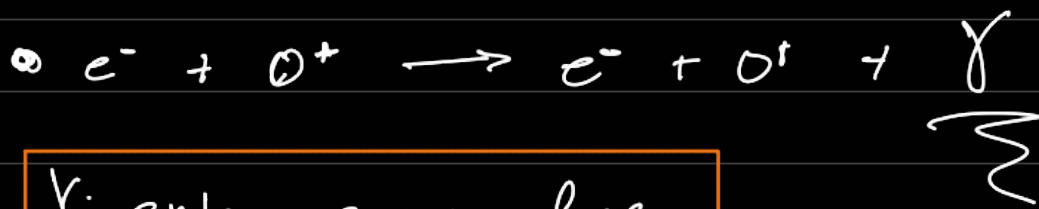
→ Amounts of O, n_O : very low

• ISM:

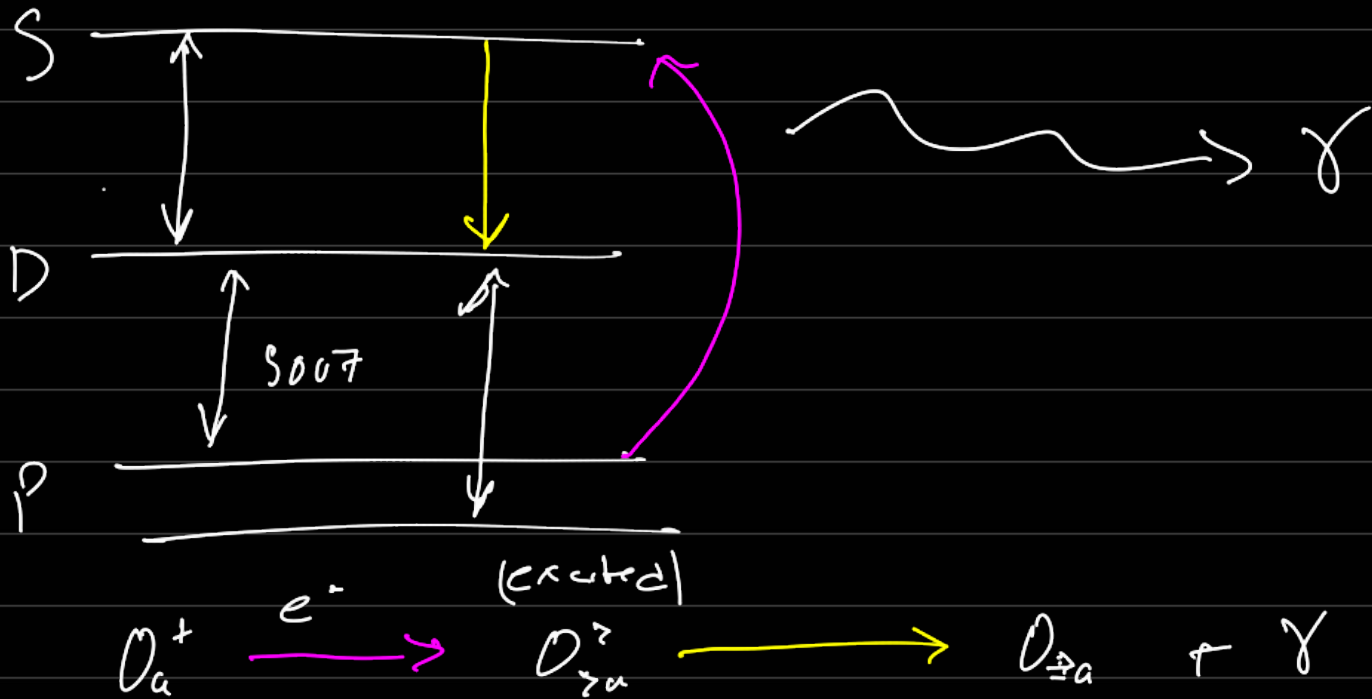
$$\left\{ \begin{array}{l} \text{O II: } (O^+) \\ \text{O III: } (O^{++}) \end{array} \right. \quad \left| \quad \text{N II: } (N^+) \right.$$

collisions/interactions in ISM

O II, O III: $T \sim 10^4 \text{ K}$ ← same for light that ionizes H II region



γ : extra energy loss



* looking at stat mechanics

$$P(n) = e^{-E_n/kT} / Z$$

$$\frac{P(n)}{P(n')} \sim e^{-(E_n - E_{n'})/kT}$$

$$I \propto P(n)$$

$$(VII) \quad \frac{\lambda_2 + \lambda_2}{\lambda_3} \rightarrow *$$

$$\frac{I_1 + I_2}{I_3} \propto e^{-\Delta E/kT} *$$

$$\begin{cases} P(n) = e^{-E_n/kT} \\ E_n = (\frac{1}{2} + n)h \\ Z = \sum_n e^{-E_n/kT} \end{cases}$$

$Z = \# \text{ protons}$

$$* D_+ \approx -13.6 \frac{Z}{n^2}$$

$$13.6 \text{ eV} = E \approx kT$$

$$13.6 \approx 10^4 K$$

